

Vijay Jawali, M.S – Data Engineer

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EXPERIENCE

Software Engineer III

Apr 2024 – Present

Walmart

Chennai, India

- Led enterprise-wide migration of Apache Spark applications from version 2.x to 3.x, ensuring seamless transition while improving performance by 25% and maintaining functionality with reduced cloud computing costs
- Automated financial reconciliation system integrating data sources to validate \$1.5B+ in daily transactions, reducing reconciliation time from 7 to 2 days and eliminating manual verification efforts
- Architected and implemented data archival solution between Cloud Storage and PostgreSQL, automating archival workflows enabling compliance with data retention policies, resulting in a 30% reduction in database storage costs
- Performed complex database migration projects, successfully transferring 25 GB of data from MySQL to PostgreSQL while ensuring zero data loss and minimal downtime
- Achieved increased operational efficiency by employing Bash and Python scripting languages to automate repetitive tasks, streamline job scheduling processes and manage clusters on GCP and WCNP

Senior Engineer

Nov 2023 – Apr 2024

Altimetrik

Bangalore, India

- Converted complex Informatica workflows to Apache Spark applications using Scala, enabling distributed computing and enhancing data processing capabilities for large-scale financial and supply chain data workloads
- Orchestrated data pipelines to load data from QuickBase to AWS S3, processed data using Apache Spark applications on AWS EMR, and loaded the processed data back to QuickBase
- Leveraged Hive databases to load and manage finance data on AWS S3, maintaining data integrity, accessibility, and compliance with financial reporting standards
- Conducted comprehensive like-to-like testing and validation, comparing traditional and cloud-based systems to ensure accurate data migration and adherence to data quality standards

Specialist Big Data Engineer

Nov 2021 – Sep 2022

Société Générale

Bangalore, India

- Enhanced transaction security on the SWIFT payment system by developing fraud detection and financial data analysis applications on a private cloud
- Upgraded financial data model to ISO20022 standards using Spark application, resulting in a 40% reduction in processing time along with an improvement in accuracy
- Conducted disaster recovery tests on Hadoop clusters to ensure continuous operation of data pipelines in the event of system failure and reduced recovery time from 12 hours to 2 hours
- Built ETL pipelines to integrate data from different source databases/APIs using Python and SQL and loaded transformed data into the warehouse for reporting purposes
- Designed and maintained Oozie orchestration flows to schedule and automate data pipelines utilizing Jenkins

Senior Data Engineer

Oct 2018 – Nov 2021

Mindtree

Bangalore, India

- Spearheaded migration of Spark modules from Mainframe to Amazon web services cloud platform, reducing data processing time by 45% while enhancing scalability
- Created real-time streaming applications using Apache Kafka, allowing faster data processing for the business
- Integrated end-to-end data pipelines with 4+ years of historical data, validating data flow between Spark jobs for cloud migration, and demonstrated data accuracy greater than 90%
- Implemented data management solutions by adapting PostgreSQL and Couchbase databases to handle and store data, harnessing the distinct features and capabilities of each platform

TECHNICAL SKILLS

Languages: Python, Scala, SQL, R

Big Data: Spark, Hadoop, Kafka, Oozie, Airflow

Databases: PostgreSQL, Hive, HBase, Oracle, Cassandra, Couchbase, Neo4j, MongoDB

Python Libraries: TensorFlow, PyTorch, Scikit-Learn, NumPy, Pandas, Matplotlib, Seaborn, pySpark, Plotly

Cloud: AWS (S3, EMR, EC2, RDS, Glue), GCP (DataProc, BigQuery, PubSub, DataFlow, DataStore), Cloudera

Development Tools: Kubernetes, GitHub, Jenkins, JIRA, VSCode, Jupyter, Maven, SBT, Confluence, IntelliJ

EDUCATION

University of Birmingham

M.Sc. Data Science, Distinction

Birmingham, United Kingdom

Sep. 2022 – Sep. 2023

Sir M. Visvesvaraya Institute of Technology

B.E. Electrical and Electronics, Distinction

Bangalore, India

Sep. 2014 – Jun 2018

CERTIFICATIONS

- * Machine Learning Professional Certificate
- * Advanced Data Science Specialization
- * Mathematics for Machine Learning
- * Data Science Professional Certificate
- * Google Data Analytics Specialization
- * Biostatistics in Public Health Specialization
- * Apache Spark with Scala
- * Hadoop Platform and Application Framework
- * Advanced Scala and Functional Programming
- * Big Data on Amazon Web Services
- * PySpark and AWS
- * Hive to Advanced Hive

PROJECTS

Multi-Document Text Summarization for Event Understanding | *Master's Thesis* 2023

- Generated comprehensive event summaries from news articles by implementing extractive and abstractive approaches with the goal of identifying the best model for summarizing CNN/DailyMail dataset articles
- Implemented and evaluated extractive summarization techniques like TF-IDF, TextRank, and Latent Semantic Analysis, utilizing unsupervised statistical and graph-based approaches
- Explored state-of-the-art abstractive summarization methods, including seq-to-seq model with pointer-generator networks and fine-tuned Transformer models like T5, BART, and LLAMA 2
- Constructed a user-friendly web interface for browsing summarized news articles, selecting events and comparing summaries generated by summarization models

Advanced Application for Infrastructure Monitoring | *NLP, OpenAI GPT-3.5, Plotly Dash* 2023

- Integrated Natural Language Processing (NLP) techniques to extract log patterns, compare against known error signatures, and interpret new logs in the context of historical failure models
- Implemented a log anomaly detection model leveraging the GPT-3.5 Turbo from OpenAI, fine-tuned on a custom dataset of over 2000 real-world Spark application logs
- Designed an intuitive web interface using Plotly Dash and Dash Components, enabling users to input raw unstructured logs and receive real-time anomaly detection, root cause analysis, and suggested fixes

Inflation Forecasting Using Time Series Analysis | *Python, Deep Learning, GCP* 2023

- Evaluated time series models by implementing statistical models (Exponential Smoothing, AR, MA, ARIMA, SARIMA) and deep learning models (RNN, LSTM, Transformer) for forecasting inflation indices
- Reduced data dimensionality to extract key features that contribute to the variance in fuel data by implementing Principal Component Analysis (PCA)
- Deployed a highly interactive Plotly Dashboard on Google Cloud Platform to explore multiple country's inflationary indices and visualize prediction in real-time

Neuroimaging Data Vault 2.0 Implementation | *Python, PostgreSQL, Data Vault* 2023

- Designed and developed a three-tier data vault architecture consisting of staging, enterprise data warehouse, and information mart layers using Python and PostgreSQL to store fNIRS medical imaging data
- Modeled hubs, links, and satellite tables in the enterprise data warehouse, adhering to data vault modeling principles and safeguarding data integrity and security through hashing techniques
- Constructed virtualized information marts using dimensional modeling (star schema) with fact and dimension tables to serve specific business needs and user groups

AWARDS AND ACHIEVEMENTS

- Received "Impact Driver" Award from Walmart for leading the automated financial reconciliation project, delivering the solution within tight deadlines while achieving the highest quality standards
- Received "Team" Award from Altimetrik for presenting POC on Gen AI, demonstrating innovation and technical leadership in emerging technologies
- Received "A-Team" Award from Mindtree six times for collaborative team spirit, exceptional teamwork, and consistent contribution to project success